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Geospatial Data Exploration using Machine Learning

Dr.K.Geetha,
Professor
Department of Information Technology,
MAM College of Engineering and Technology,
Trichy, Tamil Nadu, India
geethananthakumar @gmail.com

A.Srivani,
Assistant Professor
Department of computer science and engineering,
Mahendra collegeo of engineering - Salem campus, Tamil Nadu, India
srivani.cce@gmail.com

S.Gunasekaran,
Associate Professor,
Department of Computer Science and Engineering,
V.S.B.Engineering college,Karur ,Tamil Nadu, India.
gkarthikkumaran@gmail.com

Dr. S. Ananthi,
Assistant Professor,
Department of Computer Science and Engineering,
Sri Eshwar College of Engineering, Coimbatore, Tamil Nadu, India
ananthi.s@sece.ac.in

Mr.S.Sangeetha,
Assistant Professor,
Department of Information Technology,
Kongunadu College of Engineering and Technology,
Trichy, Tamil Nadu, India.
sangeethakncet@gmail.com
https://orcid.org/0000-0003-4661-6284

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Abstract— **Geospatial data exploration's major objective is to learn more about spatial characteristics in order to decide accommodation of students in various locations. The K-means technique, which is frequently used in geospatial analysis to find spatial patterns or hotspots of particular phenomena, is the algorithm utilised in this research. It is used to organise data points into clusters based their similarity. Urban planning, transportation analysis, and epidemiological research, among other subjects, can all benefit greatly from this kind of analysis. The process of looking into and displaying patterns, trends, and relationships within data that has been gathered and documented using geographic information is known as exploratory analysis of geolocational data. It will be possible to accommodate in this study.**

Keywords— Data exploration, Student accommodation, K-means technique.

I. INTRODUCTION

Examining and analyzing location-based data to find trends, patterns, and insights is known as geospatial data exploration. Understanding geographic features, population distribution, movement patterns, and other spatial phenomena can be improved with the use of this kind of analysis. Data collection and cleansing, data visualization on maps and other geographic representations, and statistical analysis to find relationships and correlations are all common steps in the exploratory analysis process. Urban planning, transportation, public health, and environmental monitoring are just a few of the industries that can benefit from the insights generated from exploratory analysis of geolocational data.

Large amounts of geolocational data are now accessible due to the rising popularity of smartphones and other GPS-enabled devices. This information can shed light on people's commuting habits, daily routines, and preferences. Geolocational data can be utilized in the context of student housing to comprehend how students interact with their surroundings and to inform the design of housing complexes and the supply of services and amenities. This study focuses on a university housing complex as a case study for exploratory analysis of geolocational data for student housing.

There are three steps involved in the exploration analysis. Preprocessing and Data collection, model building, and deployment come first. The dataset is downsized to the correct dimensions, unnecessary noise is eliminated, and waste data is also removed during the pre-processing stage. This is helpful for separating the geolocational data's features. The features are taken from the data after pre-processing. Model development is the process of developing a probabilistic model that best reflects the connection between the dependent and independent variables. Finding the right relationship (linear or curved) and choosing the right independent variables are the main challenges. In deployment, the process by which the document types,

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model, and general specifications are made accessible for use is known as deployment. A research work can be connected with a user application for document processing after it has been deployed.

II. RELATED WORKS

In the gathering and preparation of data, Determine the [1] data sources for geographic information on student housing, such as rental listings or information about campus transit. Acquire the data, download it [2], and then transform it into a form that may be used for analysis. Execute [3] data cleaning and preparation, which can involve clearing out duplicates, addressing missing values, and standardizing the data.

Use heatmaps and other geographic [4] visualizations to study the distribution of student housing and locate hotspots or clusters when exploring data. To find connections [5] between elements like cost, location, and amenities, use scatter plots and other visuals. To summarize the data [6] and spot patterns and trends, use descriptive statistics.

Do inferential analysis, such as regression analysis, in inferential analysis [7] and modelling to determine the variables that affect a student's choice of housing. To find spatial patterns and associations, use spatial analysis techniques such as geographically [8] weighted regression. The outcome of using these techniques demonstrates their high data [9] accuracy. Create models to simulate scenarios or anticipate future [10] trends.

Interpret the analysis's findings and describe their implications for the creation of student housing and the provision of services in [11] interpretation and discussion of results. The data points themselves serve as the input for this neural network, [12] which produces parameters for the variational distribution. Talk about the study's shortcomings [13] and offer ideas for future research possibilities. Highlight the advantages and disadvantages of the current study by contrasting the results [14] with those of earlier research.

Briefly discuss the study's main conclusions [15] and offer a concluding viewpoint on the research subject in your conclusion and in your future work. Suggest potential areas [16] for further investigation, such as extending the study's geographic scope [17] or utilizing more data sources.

The key advantage [18] of using the CNN model is that it takes advantage of the topological information contained in the data and is insensitive to fundamental changes like pivot, interpretation, and so on. However, MLP models [19] never take into account precise geographic information and are unsuitable for dealing with complicated problems.

The management of workloads in a cloud environment requires the use of APIs [20] (App programming interfaces). The AECCSs (Amazon Elastic Compute Cloud Surroundings) propose an absurd assortment of APIs that permit customers to create their own unique workloads over Amazon's register and capacity facilities.

In order to identify [21] the characteristics of the computed tomography bronchi pictures and to classify the location in the respiratory track, a convolutional auto encrypted learning algorithm is proposed in this study.

III. SYSTEM ARCHITECTURE

This module's job is to acquire geolocational data regarding student housing from a range of sources, such as websites that display available rentals, data about campus transportation, and student surveys. For subsequent examination, the gathered data is kept in a centralized database. Fig 1.a represents the architecture and the module is in charge of preparing and cleaning the obtained data in the data processing and preparation module to guarantee its quality and consistency. It also entails converting the data into an analysis-friendly format, including geographical data formats. To find patterns and relationships in the geolocational data, the data analysis and modelling module is responsible for undertaking exploratory analysis, inferential analysis, and modelling.

Using statistical methods like regression analysis, geographical analysis methods like clustering and hotspot analysis, and machine learning algorithms for predictive modelling may all be involved in this. In order to facilitate exploration and explanation, the data visualisation module is in charge of producing visualisation of the geolocational data. Heatmaps, scatter plots, bar charts, and other graphs and maps may be included in this. The module responsible for assessing the outcomes of the data analysis and modelling as well as addressing how they may affect the layout of student housing and the provision of services. Also, it entails discussing the advantages and disadvantages of the current study as well as contrasting the findings with those of earlier research.

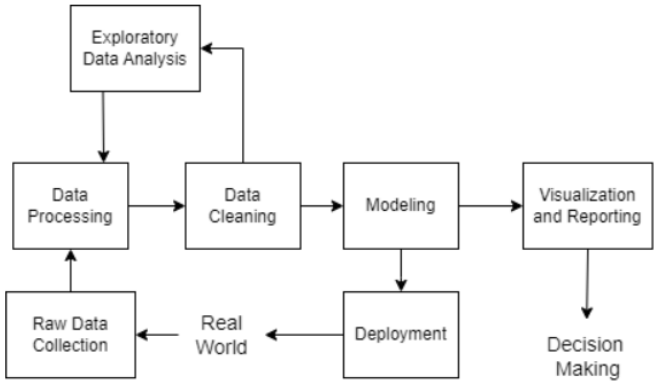

Fig. 1.a. System architecture for exploratory analysis

A. Data collection

Fig 1.b represents the collection of geospatial data can be achieved through various methods depending on the type of data and the specific application. field surveys, publicly available data, GPS and GNSS, and aerial photography. The features are taken from the data after pre-processing.

- *Data cleaning*

It assists in making sure the data is precise, consistent, and prepared for analysis. Duplicate data are eliminated, missing values are filled in, outliers are found and eliminated, and data input errors are fixed as part of the data cleaning process.

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- *Data transformation*

In order to do this, the raw data must be transformed into an analysis-ready format. Changing data types, combining data at various sizes, and re-projecting the data into a single coordinate system are a few examples of how to do this.

B. Building the model

To find patterns and relationships in the geolocational data, this module must do modelling, inferential analysis, and exploratory analysis. Using statistical methods like regression analysis, geographical analysis methods like clustering and hotspot analysis, and machine learning algorithms for predictive modelling may all be involved in this.

- *Data Visualization Module*

The geolocational data will be visualized by this module to facilitate exploration and analysis. Heatmaps, scatter plots, bar charts, and other graphs and maps may be included in this.

- *Geospatial Data Exploration*

Find out what influences students' choice of housing by using inferential analysis, such as regression analysis. To find spatial patterns and relationships, use spatial analysis techniques like regionally weighted regression. Create models to simulate scenarios or anticipate future trends.

C. Visualization and clustering

It is the responsibility of this module to interpret the outcomes of the data analysis and modelling and to go over how they might affect the design of student housing and the supply of services. Also, it entails discussing the advantages and disadvantages of the current study as well as contrasting the findings with those of earlier research.

D. K- Means algorithm

The selection of K-means clustering algorithm depends on various factors such as the size of the dataset, the desired performance, and the specific requirements of the problem at hand. K-means algorithm separates a set of observations into k clusters depending on how similar they are to the places where people do well in a certain area. It also includes picture segmentation, customer segmentation, and anomaly detection.

IV. DATASET

Plot a boxplot on the data frame to visualize the food choices dataset you prepared in the preceding step. To understand the Boxplot, go to the codebook food file. Make a note of the broad trends that the study's sampled population has been following. You should consider factors such as average activity levels (these folks will need gyms), vegetarianism or not (we should take both into account), and wealth (this affects the lifestyle of the person significantly.)

	cook	eating_out	employment	ethnic_food	exercise	fruit_day	income	on_off_campus	pay_meal_out	sports	veggies_day
0	2.0	3	3.0	1	1.0	5	5.0	1.0	2	1.0	5
1	3.0	2	2.0	4	1.0	4	4.0	1.0	4	1.0	4
2	1.0	2	3.0	5	2.0	5	6.0	2.0	3	2.0	5
3	2.0	2	3.0	5	3.0	4	6.0	1.0	2	2.0	3
4	1.0	2	2.0	4	1.0	4	6.0	1.0	4	1.0	4

Fig. 1.b. real time dataset for data analysis

V. EXPERIMENTAL OUTPUT

With the geolocational data, the geospatial data analysis may identify spatial patterns and trends, such as hotspots of high demand for student housing or regions with limited access to public transit. These patterns can shed light on the variables that affect preferences and choices for student housing. The research could include suggestions for the layout and delivery of services related to housing for students, such as the ideal site for a complex's size, facilities, and amenities. These observations can help to raise the standard and cost of on-campus housing while also enhancing the educational experience is shown in the Fig. 2.

```
C:\Users\sanjay\Documents\mini project (college)\geo_crioIDB-main>py data_api.py
```

	position.lat	position.lng	Cafes	Department Stores	Gyms
0	19.1255	72.9193	8	3	19
1	19.1235	72.9131	20	7	20
2	19.1194	72.9121	20	7	20
3	19.121	72.8854	5	3	20
4	19.1177	72.8837	6	4	20
5	19.1627	72.9329	6	2	9
6	19.096	72.9136	15	5	15
7	19.1665	72.9558	5	6	20
8	19.0781	72.9117	12	4	20
9	19.1719	72.9559	7	7	20
10	19.1628	72.8579	13	3	16
11	19.1237	72.8466	15	3	20
12	19.1462	72.8459	7	2	14
13	19.0963	72.8542	18	1	16
14	19.1019	72.8455	19	5	20
15	19.1489	72.8393	20	3	20
16	19.1489	72.839	20	3	20
17	19.125	72.9896	1	1	2
18	19.1251	72.9898	1	1	2
19	19.1256	72.9905	1	1	3

Fig. 2. Pre-processed spatial patterns

Based on historical data, demographic, and economic aspects, the analysis could lead to the creation of predictive models for demand and supply for student housing. Future demand and supply situations can be projected using these models, which can also be used to guide investment and policy choices and it is shown in the fig. 3.

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	position.lat	position.lng	Cafes	Department Stores	Gyms	Cluster
19	19.1256	72.9905	1	1	3	
0	19.1255	72.9193	8	3	19	0
1	19.1235	72.9131	20	7	20	1
2	19.1194	72.9121	20	7	20	1
3	19.121	72.8854	5	3	20	0
4	19.1177	72.8837	6	4	20	0
5	19.1627	72.9329	6	2	9	2
6	19.096	72.9136	15	5	15	1
7	19.1665	72.9558	5	6	20	0
8	19.0781	72.9117	12	4	20	0
9	19.1719	72.9559	7	7	20	0
10	19.1628	72.8579	13	3	16	1
11	19.1237	72.8466	15	3	20	1
12	19.1462	72.8459	7	2	14	0
13	19.0963	72.8542	18	1	16	1
14	19.1019	72.8455	19	5	20	1
15	19.1489	72.8393	20	3	20	1
16	19.1489	72.839	20	3	20	1
17	19.125	72.9896	1	1	2	2
18	19.1251	72.9898	1	1	2	2
19	19.1256	72.9905	1	1	3	2

Fig. 3. The output of parameters with clusters

The place you selected should be in the middle of a map with the locations delineated by colour. Check out the reference screenshot below is shown in the Fig. 4.

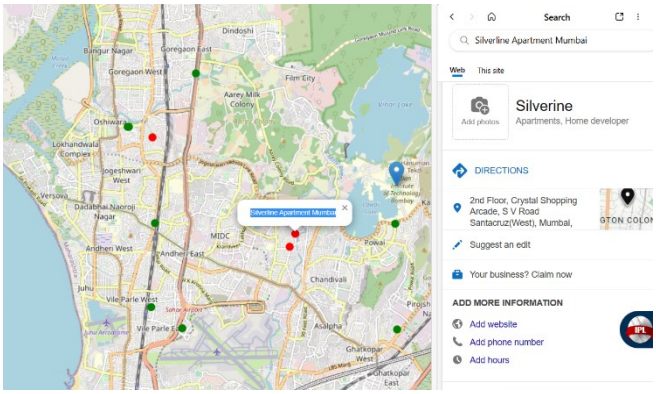

Fig. 4. The Output for Geospatial data exploration

VI. EXPERIMENTAL RESULTS FOR GEOSPATIAL DATA EXPLORATION USING ML

Metrics that gauge how well a model performs in predicting the right class labels, such as accuracy, precision, recall, and F1, can be used to gauge how accurate a classification analysis is. The experimental findings can indicate that the classification model's accuracy is, for instance, 85%, meaning that the model properly assigned 85% of the student housing units to the appropriate categories. Fig 5 represents a comparison of F1-score of different parameters to determine the accuracy.

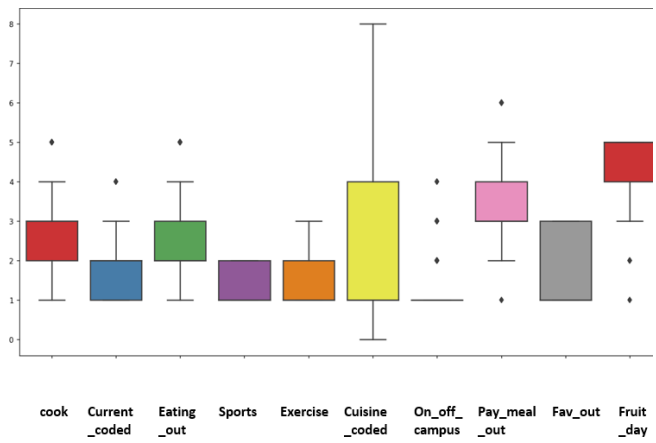

Fig. 5. Comparison for F1-score of different parameters

VII. CONCLUSION AND FUTURE WORK

In the proposed method, examining and analysing location-based data to find trends, patterns, and insights is known as Geospatial data exploration. Understanding geographic features, population distribution, movement patterns, and other spatial phenomena can be improved with the use of this kind of analysis. High performance in data exploration is achieved by combining effective data processing methods, suitable data exploration methodologies, and making use of the computational power of current hardware.

Thus, the future implementation for this kind of analysis can be used to inform decisions based on geographic data in a variety of sectors, including business, healthcare, and environmental science.

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